

Bias-Variance Tradeoff and Model Validation

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Model Selection



Model Selection



The Problem of Model Selection

- ▶ In linear regression, we often have many potential predictors.
- ▶ **Model selection** = deciding which subset of variables to include.
- ▶ Why it matters:
 - ▶ Too few predictors $\Rightarrow$ underfitting
 - ▶ Too many predictors $\Rightarrow$ overfitting
- ▶ Goal: balance complexity and prediction accuracy

What Is “Learning Theory”?

- ▶ **Statistical Learning Theory** studies how algorithms trained on a sample perform on unseen data.
- ▶ Key question: **Will the model generalize?**

$$\text{Training error} \stackrel{?}{\approx} \text{Population error}$$

- ▶ Provides tools (bounds, capacity measures) to relate sample size, model complexity, and expected prediction error.
- ▶ We emphasize *intuition* over proofs; details become crucial only when you design new algorithms.

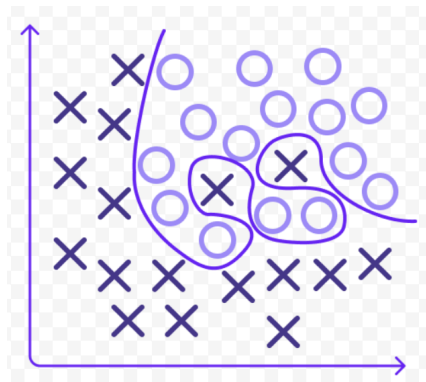
Memorization vs. Generalization

Memorization

- ▶ Perfectly fits training data (zero error).
- ▶ Fails catastrophically on new data.
- ▶ Example: lookup table of all training inputs.

Generalization

- ▶ Captures *structure* rather than noise.
- ▶ Training error is low *and* stays low on test data.
- ▶ Goal of any predictive model.

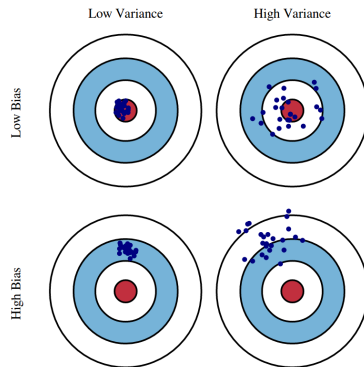


Example Scenario

- ▶ Predicting blood pressure using:
 - ▶ Age, weight, gender, cholesterol, physical activity, smoking, etc.
- ▶ Which of these predictors should we include in our model?
- ▶ What criteria can we use to choose?

The Bias–Variance Story

- ▶ **Bias**: systematic error from overly rigid models (under-fit).
- ▶ **Variance**: sensitivity to idiosyncrasies of training sample (over-fit).
- ▶ **Sweet spot** minimizes expected prediction error.
- ▶ Take-home: adding complexity helps until variance explodes.



Model Selection Goals:

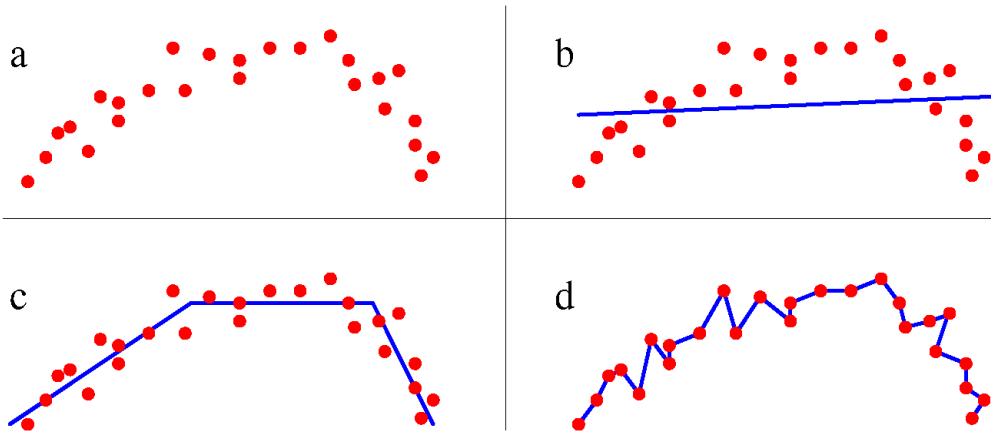
To make reasonable predictions or estimations, we need

- ▶ accuracy → on average, what we estimate is equal to what we expect (e.g., the predicted blood pressure is equal to the average blood pressure for all groups of people).
 - ▶ The predicted blood pressure equals the average blood pressure for all groups of people.
- ▶ precision → small variation in prediction/estimation
 - ▶ The predicted blood pressure is close to the true blood pressure for all groups of people.

How Do We Spot Overfitting?

- ▶ **Gap** between training and validation error grows.
- ▶ Highly erratic predictions for small data perturbations.
- ▶ Model relies on “spurious” variables (e.g., patient ID).
- ▶ Visualization: training curve vs. validation curve.

Underfitting vs. Overfitting



Underfitting vs. Overfitting

- ▶ Underfitting occurs when important regressors are left out of the model. Costs:
 - ▶ deficient models (i.e., missing patterns).
 - ▶ misinterpretations of variable relationships.
- ▶ Overfitting occurs when all important regressors are in the model, but some unimportant ones are, too. Costs:
 - ▶ unneeded complexity and increased variance of the predicted values.
 - ▶ widened confidence and prediction intervals.

Practical Tools for Better Generalization

Cross-Validation

- ▶ k-fold, LOOCV, stratified CV
- ▶ Provides unbiased estimate of out-of-sample error.

Regularization

- ▶ Penalize model complexity (ridge, lasso, dropout).
- ▶ Shrinks coefficients or removes them entirely.

Early Stopping

- ▶ Monitor validation loss during training (NNs, boosting).
- ▶ Stop when loss stops improving.

Ensembling

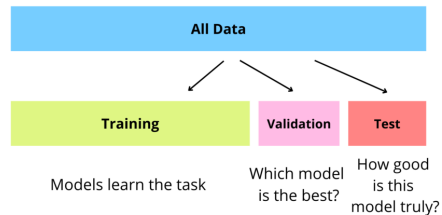
- ▶ Combine many weak learners (bagging, random forests).
- ▶ Averages out variance while retaining flexibility.

Evaluation Criteria

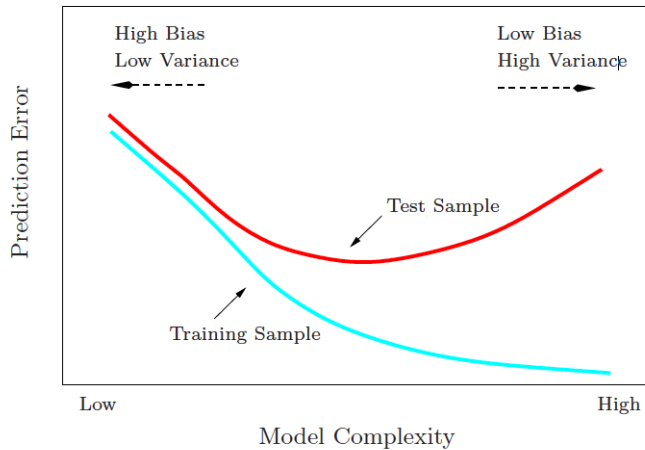
- ▶ **Training Error:** How well the model fits to the data used to estimate the parameters.
- ▶ **Test Error:** How well it predicts *test data*, i.e., data not used in the estimation of the parameters.
- ▶ **Cross-Validation:** procedure of cycling through the data, estimating the performance on disjoint groups of test data.
- ▶ **Information Criteria:**
 - ▶ AIC: balances fit and complexity
 - ▶ BIC: stronger penalty for complexity

Introduction to Validation

- ▶ Validation involves splitting the data into training, validation, and test sets.
- ▶ **Training Set:** The portion of data used to train the model, i.e., estimate $\hat{\beta}$.
- ▶ **Validation Set:** The portion of data used to tune model parameters and prevent overfitting, i.e., to tell us which subset of predictors to use in our model.
- ▶ **Test Set:** The portion of data used to assess the final performance of the model after training and validation.



Test vs Training Error



Introduction to Cross-Validation

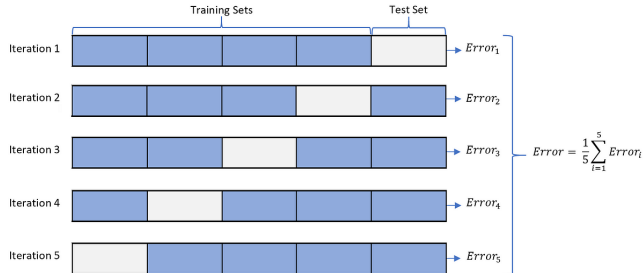
- ▶ **Definition:** Cross-validation (CV) is a statistical method used to estimate the performance of machine learning models.
- ▶ **Purpose:** It helps in assessing how a model generalizes to an independent dataset.
- ▶ Why Use Cross-Validation?
 - ▶ **Prevents Overfitting:** Ensures model's robustness.
 - ▶ **Provides Reliable Estimates:** Offers a very good estimate of model performance.
 - ▶ **Optimizes Hyperparameters:** Helps choose parameters that can't be estimated (tuning parameters, e.g., the number of variables in a model).
 - ▶ **Accuracy Measures:** Any type can be used.

Basic Concepts

- ▶ **Divide Data:** Split data into training and validation (or test) sets.
- ▶ **Multiple Iterations:** Perform the split multiple times.
- ▶ **Aggregate Results:** Average the performance metrics.
- ▶ Types of CV:
 - ▶ K-Fold Cross-Validation
 - ▶ Leave-One-Out Cross-Validation (LOO)
 - ▶ Stratified K-Fold Cross-Validation
 - ▶ Time Series Split (for time-dependent data)

Cross-Validation Details

- ▶ Steps of K -fold cross-validation:
 1. Split data into K subsets (folds).
 2. Train/Estimate on $K-1$ folds and validate on the remaining fold.
 3. Repeat K times.
 4. Average the results.
- ▶ With LOO, each person is their own fold.



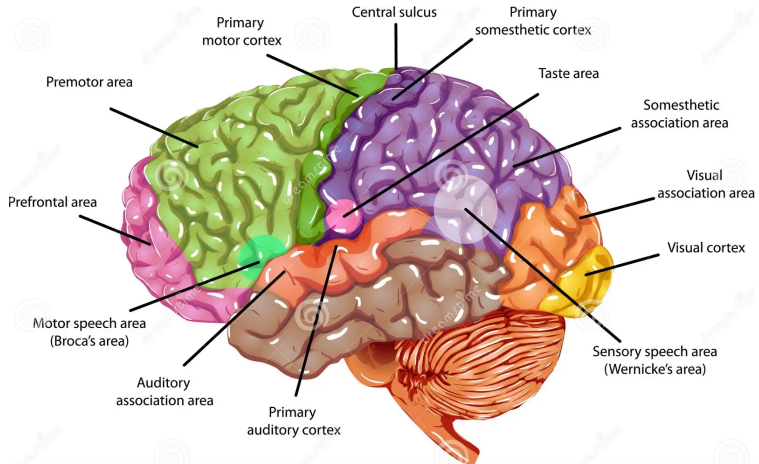
Multiple Linear Regression

- ▶ One dependent variable Y .
- ▶ p independent variables:
 - ▶ X_j = proportion of voxels damaged in Region Of Interest (ROI) j .
- ▶ The model is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon$$

or $E(\mathbf{Y}) = \mathbf{X}'\boldsymbol{\beta}$.

- ▶ β_0 : intercept;
- ▶ $\beta_j, j = 1, \cdots, p$: regression coefficients.
- ▶ $E(\epsilon) = 0$ and $var(\epsilon) = \sigma^2$



What X to use?

- ▶ Recall, that X_j = proportion of voxels damaged in ROI j .
- ▶ How do we define the regions? Some options:
 1. **Harvard-Oxford Atlas**, Number of ROIs (i.e., p): 48
 2. **Automated Anatomical Labeling (AAL) Atlas**, Number of ROIs: 116
 3. **Brainnetome Atlas**, Number of ROIs: 246
 4. **Schaefer Atlas**, Number of ROIs: Varies (100, 200, 300, 400, 500, 600, 700, 800, 900, 1000)
 5. **Voxel level**, $p > 10^5$.

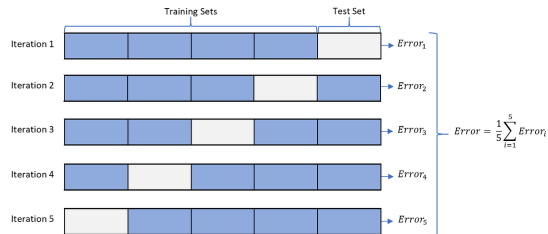
Example: Predicting AQ with brain images (with 5-fold CV)

1. Put the people into 5 groups.

Let $G_i = k$ if person i is in group k

2. Then for $k = 1, 2, 3, 4, 5$:

- 2.1 Use $k - 1$ folds to fit a linear model to every brain atlas.
- 2.2 Using the fitted linear models, predict AQ for every person in the k th group, i.e., the test fold.
- 2.3 Get the total error for the test group for each brain atlas. For brain atlas 'a' this would be:



$$\widehat{SSE}_k^a = \sum_{i; G_i=k} (Y_i - \hat{Y}_i^a)^2$$

where $\hat{Y}_i^a$ is the predicted AQ when brain atlas 'a' is used.

Example: Predicting AQ with brain images (cont.)

3. Get the total error over all folds:

$$\widehat{SSE}^a = \sum_{k=1}^5 \widehat{SSE}_k^a, \quad \widehat{SSE}^b = \sum_{k=1}^5 \widehat{SSE}_k^b, \quad \widehat{SSE}^c = \sum_{k=1}^5 \widehat{SSE}_k^c, \quad \text{etc.}$$

4. Use the ROI Options that has the lowest $\widehat{SSE}_j$.

This will give us an estimate of which ROI has the best predictive ability. No need to use forward/backward selection or model fit criteria.

Why the CV error of the “winner” is biased

- ▶ Suppose $\widehat{SSE}_a$ was the smallest.
- ▶ Think of CV error as $\widehat{SSE}_j = SSE_j + \varepsilon_j$ with $\mathbb{E}[\varepsilon_j] = 0$.
- ▶ However, we searched over many models and picked the ε_{j^*} with the best SSE .
- ▶ Selection makes ε_{j^*} tend to be **negative** (lucky noise).

$$\mathbb{E}[\widehat{SSE}_{j^*}] < SSE_{j^*}.$$

- ▶ This is the **winner's curse** / selection-induced optimism:
 - ▶ The more candidates you try, the larger the expected optimism.
 - ▶ Reusing the *same* CV folds for both tuning and reporting amplifies it.

How to get an honest performance estimate

► **Nested CV (recommended):**

1. **Outer** split into K_{outer} folds.
2. On each outer train set, run **inner** CV to tune/choose the model.
3. Refit the chosen model on the full outer train set; evaluate on the outer hold-out fold.
4. Aggregate outer-fold errors $\Rightarrow$ **unbiased** estimate for the selection procedure.

► **External test set:** If data permit, hold out a test set that is never touched until the very end.

► **One-standard-error rule:** Choose the *simplest* model whose CV error is within 1 SE of the minimum to reduce over-tuning.

Common Pitfalls

Common Pitfalls and Misuses

- ▶ **Data Leakage:** Ensure validation data do not influence training data.
 - ▶ e.g., centering the outcome and covariates at once.
- ▶ **Improper Splitting:** Avoid splitting time series data randomly (only want to predict forward).
- ▶ **Overlapping Data:** Ensure folds are mutually exclusive.

CV Takeaways

- ▶ **Proper Data Splitting:** Maintain clear boundaries between training, validation, and test sets.
- ▶ **Avoid Data Leakage:** Ensure no information from the test set influences the training process.
- ▶ **Avoid Data Overlap:** When working with grouped data (e.g., multiple visits per patient), ensure that groups are kept intact during cross-validation to prevent data leakage.
- ▶ **Respect the Temporal Order:** When working with time series or temporally ordered data, it is crucial to maintain the chronological order to avoid data leakage.
- ▶ **Evaluate Performance:** Use appropriate metrics and validation techniques to ensure reliable model performance assessment.

Google Flu Trends (2008–2014)

- ▶ **Idea:** Predict U.S. influenza levels by mining billions of Google search queries.
- ▶ **Early success:** Cross-validation on historical CDC data looked stellar.
- ▶ **Overfitting outcome:** Model keyed on ephemeral search spikes (media coverage, unrelated “flu” terms) and over-predicted the 2013 season by $\sim 140\%$.
- ▶ **Lesson:** Behavioral data streams drift quickly—continual re-training and ground-truth checks are essential.

Genomic “Sepsis Signature” in the ICU (2016)

- ▶ **Claim:** A 20-gene micro-array achieved 92% accuracy distinguishing bacterial vs. viral sepsis.
- ▶ **Problem:** Discovery and “validation” cohorts shared the same batch effects and platform.
- ▶ **Aftermath:** Independent hospitals saw accuracy drop to the low 60% range.
- ▶ **Lesson:** High-dimensional $p \gg n$ studies require external, multi-site validation and batch correction.

IBM Watson for Oncology (2015–2018)

- ▶ **Goal:** Recommend cancer treatments worldwide using cases from Memorial Sloan Kettering.
- ▶ **Issue:** Recommendations ignored local formularies, insurance rules, and rare pathology sub-types not seen in training.
- ▶ **Result:** Several hospitals withdrew after inconsistent or unsafe suggestions.
- ▶ **Lesson:** Elite single-site data can be parochial—robust generalization needs diverse, representative training sets.

Epic's Proprietary Sepsis Model (2017–2021)

- ▶ **Advertised performance:** AUROC ≈ 0.80 on internal tests.
- ▶ **Independent audit:** Three Midwestern hospitals found sensitivity $< 40\%$; many true sepsis cases missed.
- ▶ **Contributing factors:** Unknown feature weights, static thresholds, population shift.
- ▶ **Lesson:** Vendor black-box models are not exempt from overfitting—demand transparent, local validation.

COVID-19 Chest X-ray Classifiers (Spring 2020)

- ▶ **Competition hype:** CNNs reported $>95\%$ sensitivity distinguishing COVID pneumonia from “normal” lungs.
- ▶ **Confounder:** Datasets mixed adult COVID images with pediatric controls—models learned age-related anatomy, not pathology.
- ▶ **Reality check:** Balanced, age-matched test sets drove accuracy toward chance.
- ▶ **Lesson:** Spurious correlations lurk in hastily compiled data; stratify splits by site, scanner, and demographics.

Key Themes Across the Anecdotes

- ▶ Spurious signals and concept drift undermine apparent accuracy.
- ▶ Narrow, single-context training data rarely generalize broadly.
- ▶ Transparent external validation and continuous monitoring are non-negotiable safeguards.